

PURERUA, New Zealand

WMO: 930230

Lat: 35.125S

Lon: 174.017E

Elev: 87

StdP: 100.29

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	7.1	2.8	4.7	7.8	3.8	5.0	8.6	16.0	12.8	13.5	13.8	2.9	200	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.9	24.0	19.7	23.2	19.3	22.5	19.0	21.5	22.5	20.8	22.0	20.2	21.6	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	15.9	22.0	20.4	15.2	21.5	19.7	14.6	21.1	62.9	22.7	60.6	22.1	58.5	21.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.1	9.3	7.8	DB	4.5	26.3	1.0	1.1	3.8	27.1	3.2	27.8	2.7	28.4	2.0	29.2	
			WB	3.5	22.4	0.9	0.9	2.8	23.1	2.3	23.6	1.8	24.2	1.1	24.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	18.9	19.6	18.6	16.9	14.9	12.9	12.1	12.2	13.1	14.1	15.5	17.6
	DBStd	3.09	1.57	1.48	1.53	1.71	1.98	1.97	1.77	1.60	1.58	1.53	1.70	1.59
	HDD10.0	8	0	0	0	0	0	2	3	2	1	0	0	0
	HDD18.3	1139	12	4	15	49	107	162	192	191	157	132	86	33
	CDD10.0	2023	276	269	268	207	153	90	69	70	94	126	166	236
	CDD18.3	112	30	40	24	6	1	0	0	0	0	0	1	10
	CDH23.3	64	23	29	7	1	0	0	0	0	0	0	1	4
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	3.3	3.3	3.3	3.4	3.1	3.1	3.2	3.5	3.4	3.3	3.3	3.3	3.2
(15) Precipitation	PrecAvg	1463	95	107	112	132	149	167	179	158	121	98	84	103
	PrecMax	1877	490	275	336	380	249	341	390	277	308	212	237	240
	PrecMin	1107	10	29	4	6	8	60	32	63	44	40	10	0
	PrecStd	256	100	66	94	86	64	74	87	51	57	43	59	62
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	25.5	24.2	22.4	20.3	18.4	17.2	17.1	18.7	20.3	22.4	23.9
		MCWB	19.6	20.7	19.4	19.2	17.8	16.8	15.2	14.2	14.7	15.2	17.0	18.3
	2%	DB	23.9	24.2	23.1	21.3	19.6	17.5	16.5	16.3	17.5	18.8	20.8	22.4
		MCWB	19.3	20.3	19.1	18.5	17.3	15.9	14.4	13.8	14.4	14.9	16.5	18.2
	5%	DB	22.9	23.4	22.3	20.6	18.9	16.9	15.8	15.7	16.8	18.0	19.8	21.5
		MCWB	19.4	19.8	18.8	18.1	17.0	15.2	13.8	13.5	14.2	14.7	16.2	18.0
	10%	DB	22.1	22.6	21.7	20.0	18.3	16.3	15.2	15.2	16.1	17.3	18.9	20.7
		MCWB	18.8	19.4	18.6	17.6	16.4	14.6	13.2	13.1	13.8	14.5	15.9	17.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	22.6	21.6	21.1	19.1	17.6	16.2	15.8	16.2	17.0	18.9	20.8
		MCDB	22.9	23.7	22.3	21.5	19.4	18.0	16.8	16.2	17.1	18.3	20.1	22.0
	2%	WB	21.0	21.7	20.8	19.9	18.3	16.6	15.2	15.0	15.6	16.3	18.1	19.9
		MCDB	22.4	22.8	21.8	20.5	19.0	17.1	15.9	15.7	16.5	17.5	19.3	20.9
	5%	WB	20.3	21.1	20.1	19.1	17.6	15.8	14.5	14.3	15.1	15.8	17.4	19.2
		MCDB	21.8	22.3	21.5	20.0	18.5	16.5	15.3	15.2	16.1	17.0	18.6	20.3
	10%	WB	19.7	20.5	19.3	18.2	16.8	15.0	13.7	13.5	14.5	15.3	16.7	18.6
		MCDB	21.3	21.9	21.1	19.5	18.0	16.0	14.8	14.7	15.7	16.6	18.2	19.8
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	5.9	6.0	5.8	5.6	5.4	5.4	5.7	6.0	6.4	6.4	6.0
		MCDBR	7.1	6.8	6.6	6.0	5.4	5.0	5.3	5.8	6.3	7.0	7.5	7.1
		MCWBR	4.1	4.0	3.8	3.8	3.9	3.8	3.9	4.0	4.3	4.5	4.6	4.4
	5% WB	MCDBR	5.2	4.6	4.7	4.1	4.1	3.9	4.0	4.4	4.8	5.3	5.7	4.9
		MCWBR	3.7	3.2	3.3	3.3	3.5	3.7	3.5	3.8	3.8	4.1	4.2	3.6
(40) Clear-Sky Solar Irradiance	taub		0.343	0.341	0.329	0.316	0.310	0.304	0.307	0.309	0.332	0.334	0.333	0.341
	taud		2.576	2.614	2.637	2.667	2.655	2.654	2.637	2.627	2.533	2.533	2.566	2.587
	Ebn at Noon		993	974	949	907	858	836	851	900	930	969	997	999
	Edh at Noon		105	98	89	76	68	64	68	78	97	105	106	105
(44) All-Sky Solar Radiation	RadAvg		6.81	5.97	4.91	3.61	2.59	2.13	2.31	3.15	4.26	5.49	6.28	6.50
	RadStd		0.43	0.41	0.32	0.23	0.14	0.13	0.14	0.18	0.20	0.34	0.43	0.56

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page